

Designing Thinking Machines

The Future of AI with Graphs-Based AI Agents

WorkshopsTage 2026

OST – Ostschweizer Fachhochschule, Rapperswil SG
8 September 2026



Who we are - iCoSys

- Part of **HES-SO**, located in Fribourg.
- **Interdisciplinary** centre of **excellence** and national reference institute in applied AI and complex systems.
- Develops **impactful** joint projects with **companies** and the **public** sector.



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Wayfarer Event Planning Assistant

From business content to evidence-based decisions

Document pack



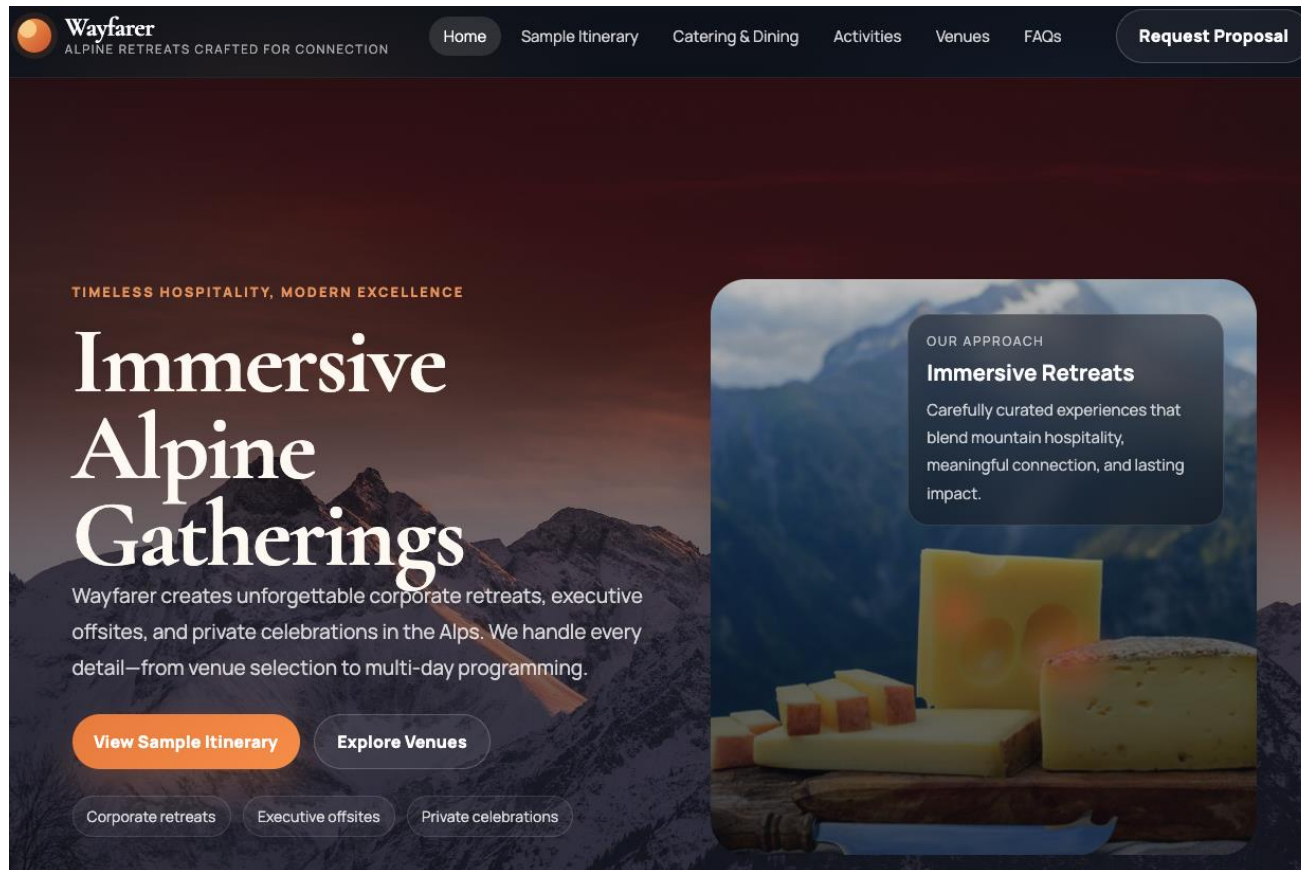
Wayfarer Event Website



Email exchange



Fondue Recipe



https://federicab03.github.io/ARGO_Usease/



Goal: Build an assistant that helps event organizers evaluate requests and make operational decisions using information from business documents and webpages, supported by clear, traceable evidence.



User request

“Does Wayfarer have a venue that meets the requirements in the email exchange?”

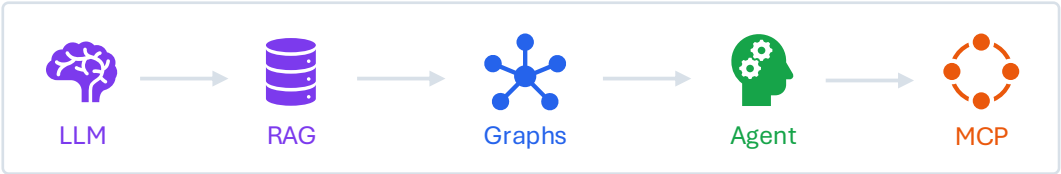


Answer with Evidence:

- 1) **Decision** (Suitable / Conditional / Not Suitable)
- 2) **Evidence** (Requirements, venue details, ...)
- 3) **Source** (Document snippets, emails)

Workshop Timeline

We build the event contract assistant by adding one capability at a time.



09:05 – 10:15	Part 01: Foundations	• LLM basics, first PydanticAI app • RAG introduction
09:05 – 10:15	Break	
10:30 – 11:30	Part 02: GraphRAG	• Structural Graph • Sematic Graph
11:30 – 13:30	Lunch	
13:30 – 14:55	Part 03: Agents & Harness	• Simple tool calling, document tools
14:55 – 15:10	Break	
15:10 – 16: 00	Part 03: MCP, Q&A	• Connect tools through MCP • Guardrails



LLM



RAG



Graphs



Agent



MCP

Workshop Repository

Github: https://github.com/ovaccarelli/workshoptage_2026

workshoptage_2026/

├ data/

├ exercises/

│ ├── part_01/

│ ├── part_02/

│ ├── part_03/

│ └ part_04/

├ presentation/

← this presentation

├ solutions/

← all the solutions to the exercises

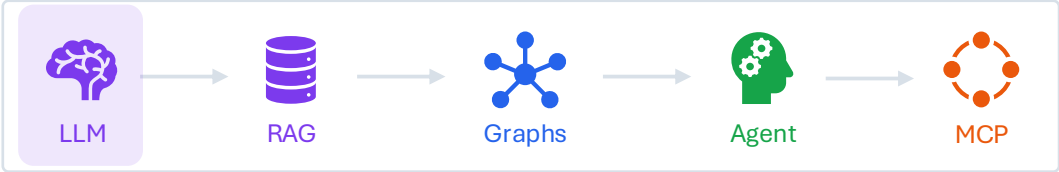
├ src/

← utils

└ README.md

What is an LLM?

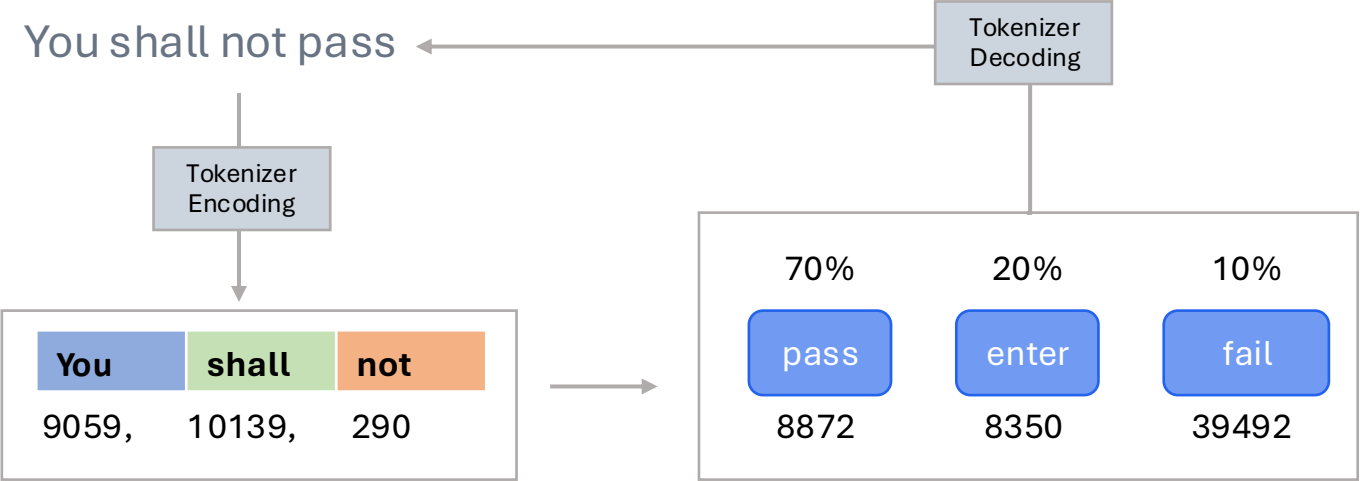
AI system trained on vast amounts of text that can understand context and generate human-like language.



Core Idea

- 1 Context in
- 2 Tokenize
- 3 Predict (next-token probabilities)
- 4 Decode & repeat

You shall not pass



Good at



Summarization/
Translation



Question
Answering



Text
Generation



Reasoning
over Context

Limits



Hallucinations



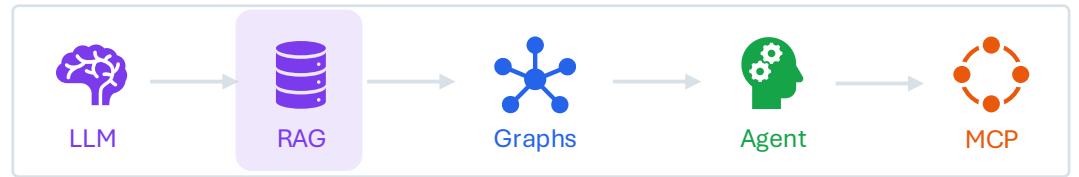
Context
Limit



No native access
to external data



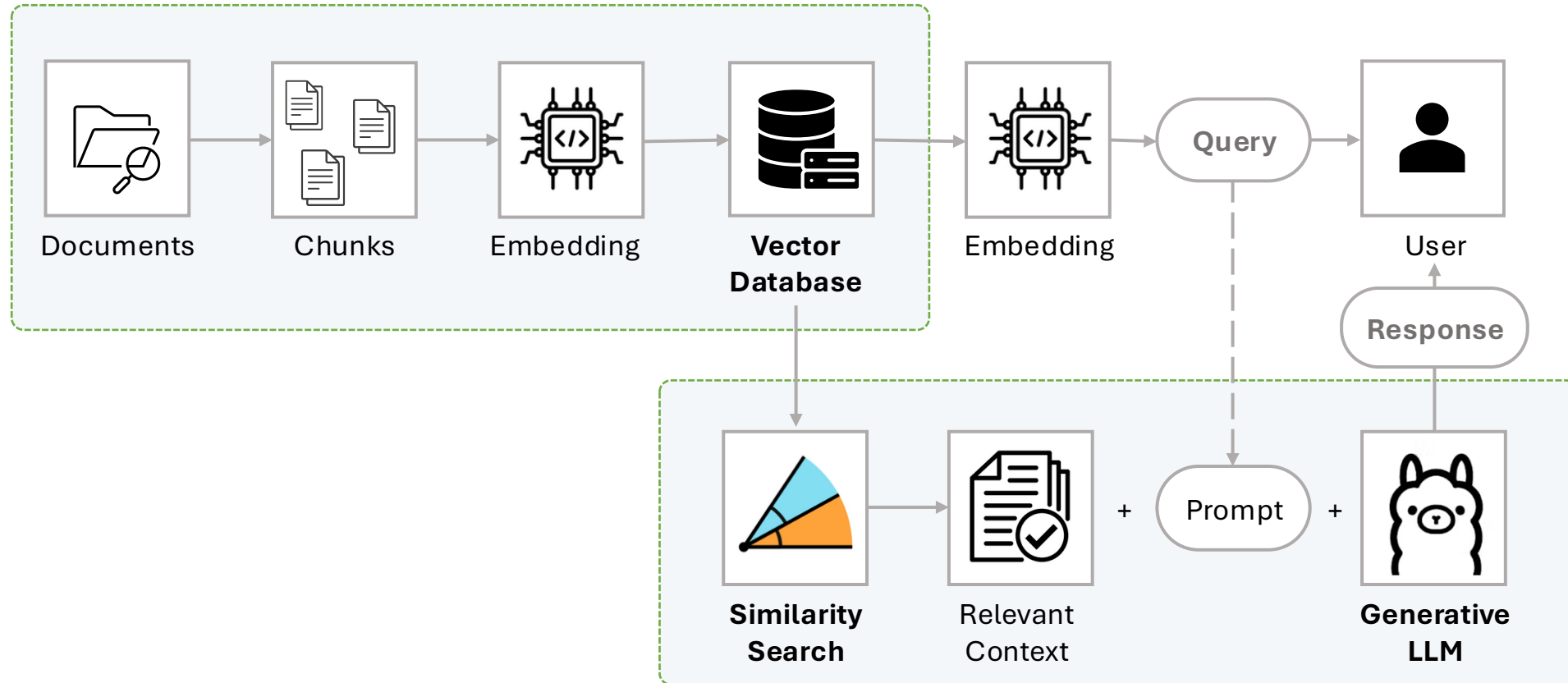
Need tools
to act



RAG: Retrieval Augmented Generation

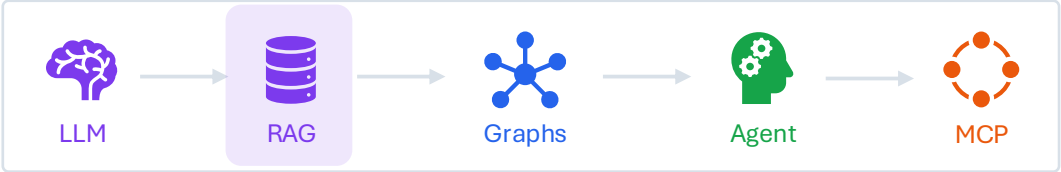
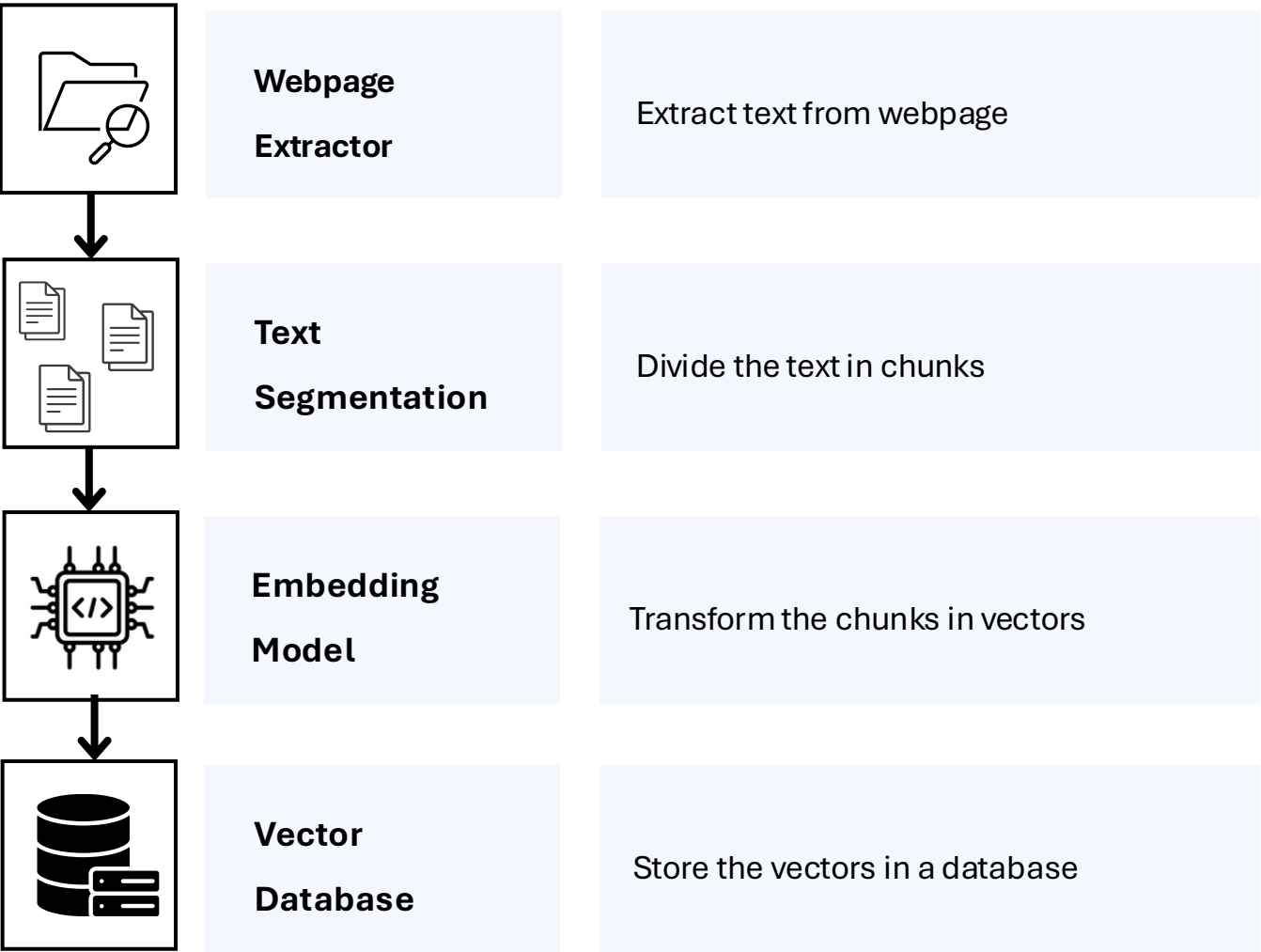
The model answers using retrieved document chunks instead of guessing.

Offline Data Preprocessing



Online Semantic Retrieval

RAG: Offline Data Preprocessing



Wayfarer

Source: https://federicab03.github.io/ARGO_Usecase/index.html

Alpine retreats crafted for connection

Chunk 0

Immersive Alpine Gatherings

Source: https://federicab03.github.io/ARGO_Usecase/index.html

Timeless hospitality, modern excellence

Wayfarer creates unforgettable corporate retreats, executive offsites, and private celebrations in the Alps. We handle every detail—from venue selection to multi-day programming.

- 14+ Years of experience
- 2 Curated venues
- 20-150 Guest capacity range

Chunk 1

Every gathering is uniquely yours

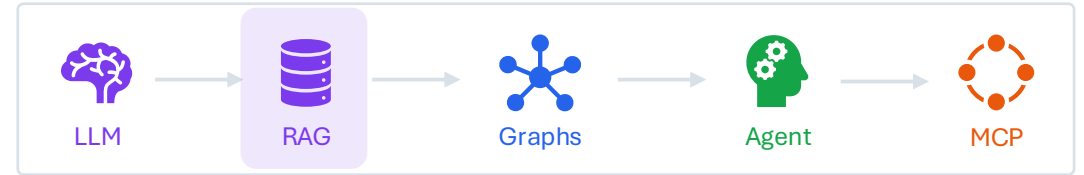
We craft experiences tailored to your group's goals and culture. Whether you're aligning teams, celebrating milestones, or exploring strategic futures, we design the retreat to fit your vision.

Our curated venues, professional catering, and personalized service all converge to create a gathering that resonates with authenticity and purpose.

Tailored experiences for every group

- **Catering & Dining** Local alpine specialties and international cuisines.
- **Activities & Entertainment** Guided hikes, team-building exercises, and seasonal experiences.
- **Venue & Logistics** Two distinct venues, each offering unique amenities and capabilities.

RAG: Online Semantic Retrieval

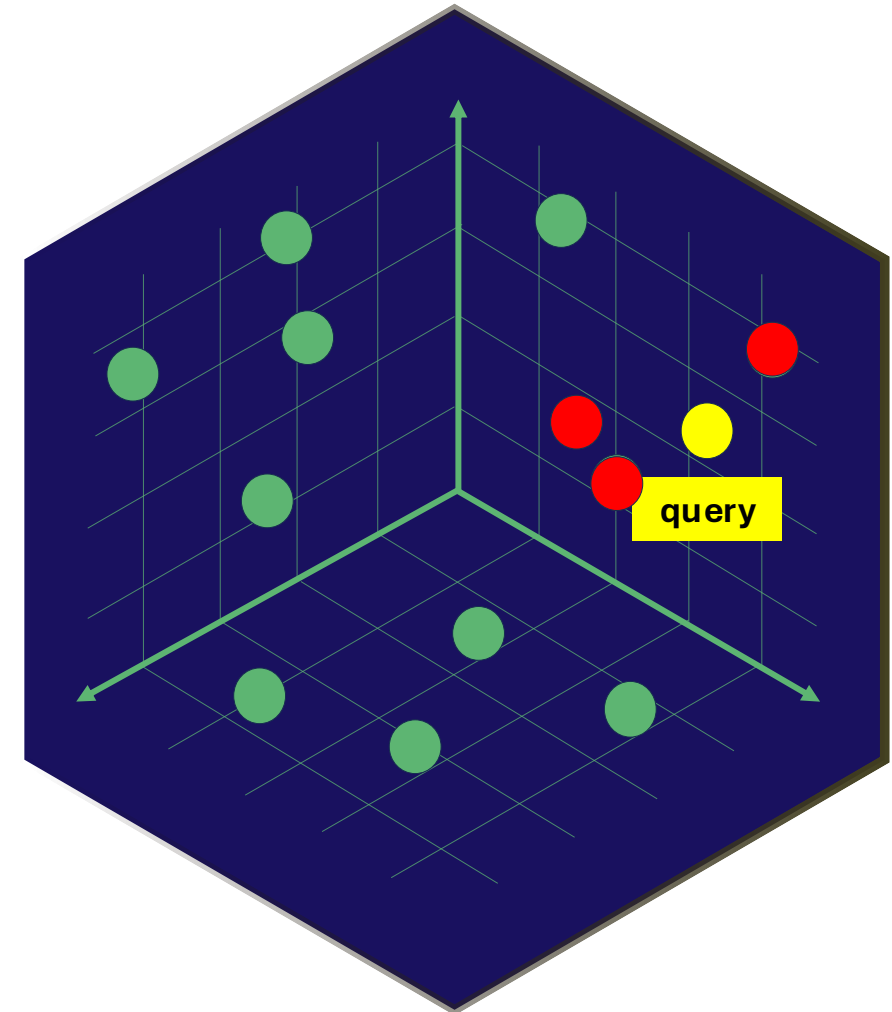
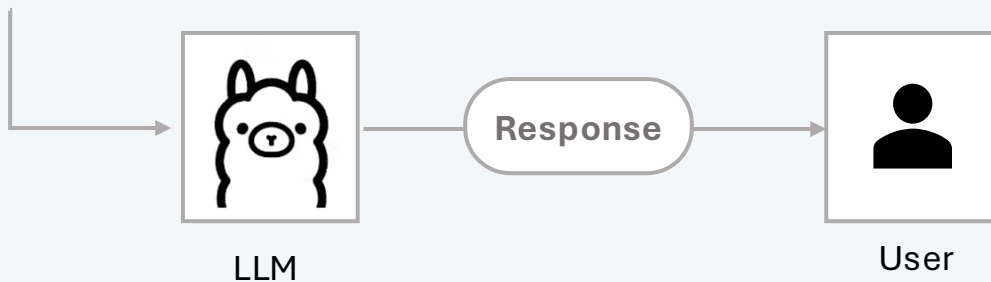


Query : also transform into a vector.

We search the K=3 chunks "similar" to the query.

• **Metric:** Cosine similarity
$$\text{Cos_sim}(\mathbf{q}, \mathbf{v}_i) = \frac{\mathbf{q} \cdot \mathbf{v}_i}{\|\mathbf{q}\| \|\mathbf{v}_i\|}$$

System Prompt + User prompt + Relevant Context

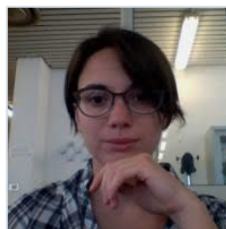


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BREAK



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Euler's Bridges of Königsberg (1736)

From a city map to a graph: the origin story

The question

Can you cross each bridge exactly once? (Eulerian Trail)

Euler realized that the exact geography did not matter. What mattered was only the **pattern of connections**.

Graph model

Nodes = Landmasses
Edges = Bridges

This abstraction gave birth to **graph theory**!
A graph removes the details and keeps the relationships.

Graph intuition

Today, we use the same idea for **knowledge**:
documents, entities become **nodes**, and their links become **edges**.



LLM



RAG



Graphs



Agent

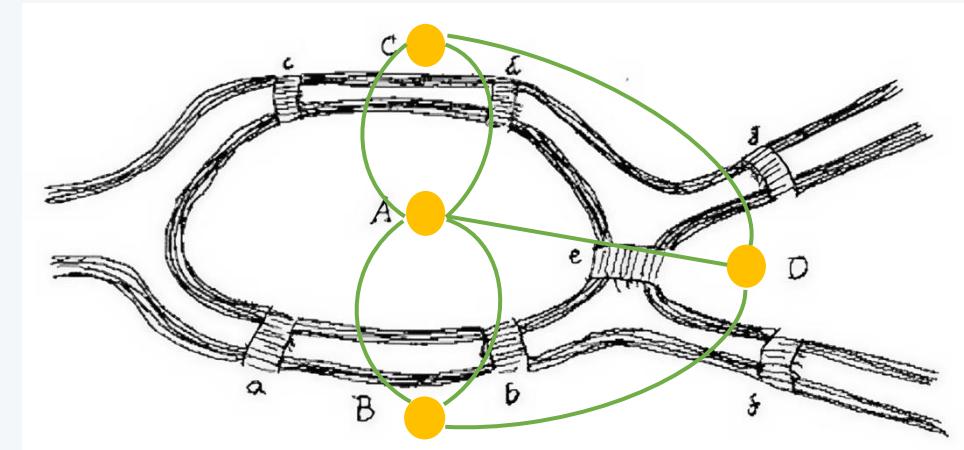


MCP

The seven bridges of Königsberg

Abstraction step:

Real map → graph (nodes/edges)



Euler's insight:

A trail exists if there are 0 (Eulerian circuit) or 2 vertices of odd degree

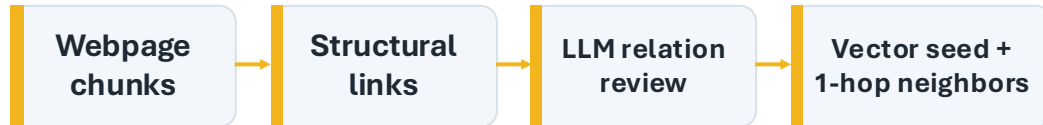
Two Graph-RAG implementation options

Structural context expansion or semantic reasoning in Neo4j

OPTION 1

Structural graph + LLM review

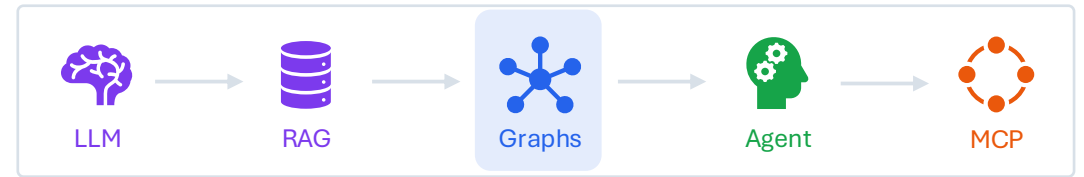
Keep the webpage hierarchy, then use the LLM to correct structural relationships.



Best fit

Reliable local context with a simpler pipeline. The graph can run from JSON, with Neo4j as an optional store.

Trade-off: relationships follow document structure, so reasoning stays close to the retrieved section.



OPTION 2

Semantic knowledge graph in Neo4j

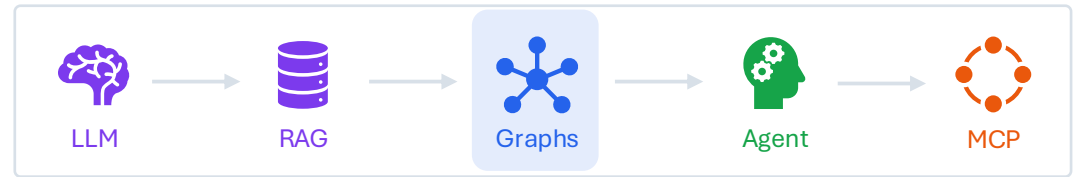
Use an LLM pipeline to extract entities and relationships, then query the graph through Cypher.



Best fit

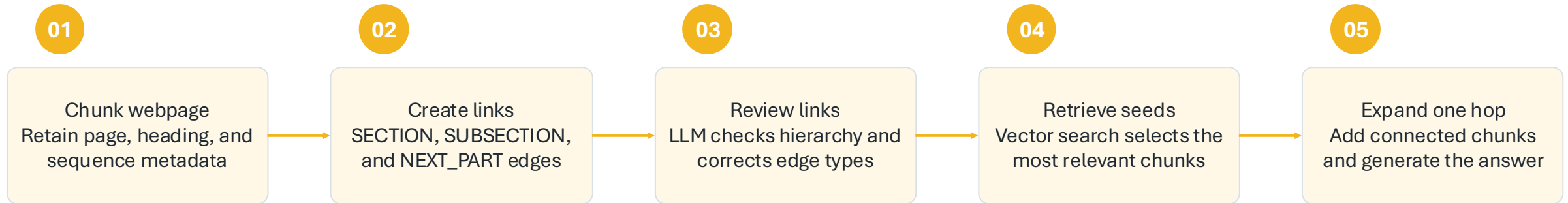
Questions that connect venues, constraints, requirements, and other entities across the source content.

Trade-off: richer semantics require Neo4j, an extraction schema, and more validation of generated relationships.



Option 1: structural graph with one-hop retrieval

The graph preserves webpage hierarchy and extends vector results with nearby context



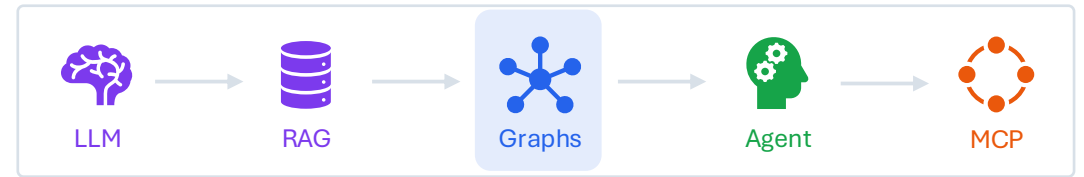
What the graph stores

Nodes represent webpage chunks. Edges preserve section membership, subsection membership, and continuation between split chunks.

Retrieval behavior

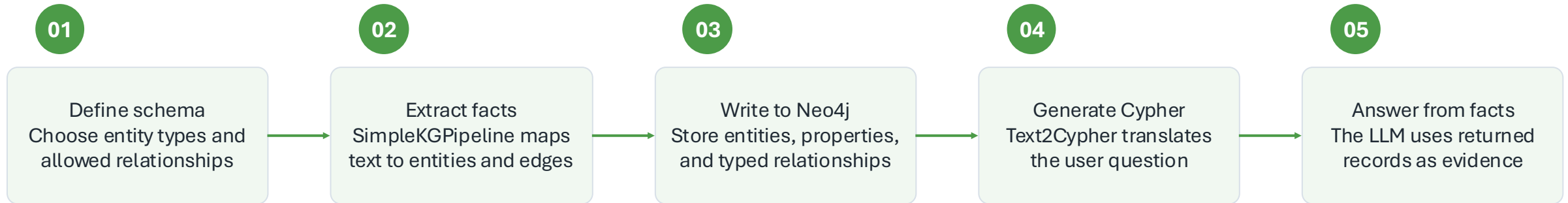
Semantic similarity finds the starting chunks. Graph traversal adds every direct neighbor, giving the LLM the local context around each match.

Use this option when page structure carries meaning and the main goal is dependable context expansion with limited infrastructure.



Option 2: semantic Graph-RAG in Neo4j

The LLM converts source text into entities and typed relationships for graph queries



Example semantic model

Venue connects to capacity, accessibility, accommodation, and event requirements. Relationships carry the meaning that structural adjacency cannot express.

Validation points

Constrain the schema, inspect extracted relationships, restrict generated Cypher to read-only queries, and answer only from returned records.

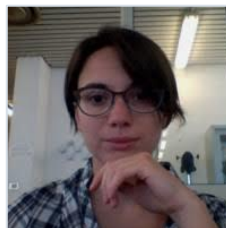
Use this option when questions must connect facts across sections or sources and the added Neo4j setup is justified.

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BREAK



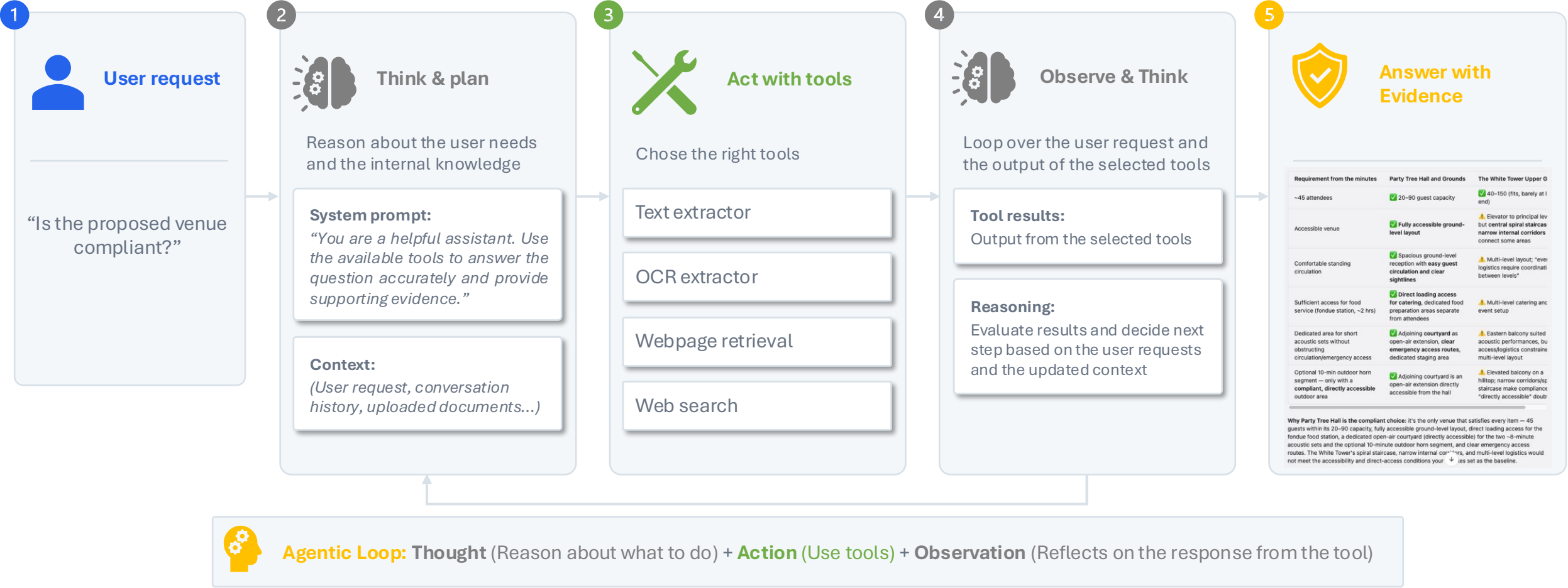
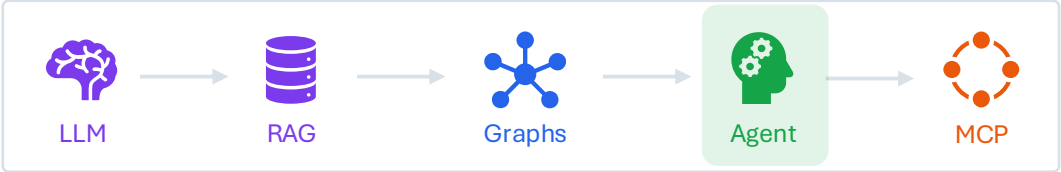
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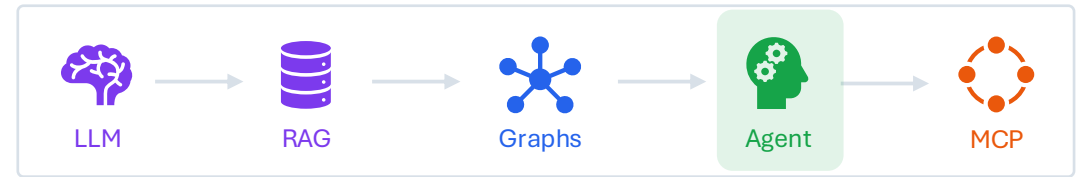
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Agent

LLM-based system that can reason, choose tools, and act to complete a task.



Tools



Tools give the agent capabilities beyond the model

Definition

- A tool is a function or service that the agent can call to get information or perform an action.
- The agent decides when to call which tool based on the user's request and available tools.

Good tools

- Have clear, specific name (e.g., `get_current_time`)
- Have clear description (When to use, what it does, what it returns)
- Have a well-defined input & output schemes (Parameters the model can fill)
- Return structured, useful results (Help the agent to reason)

Think & plan

Based on the user request and available tools the model decides to call a tool

Tool calling

The model call the selected tools

Tool execution

The system runs the tool and returns the result

Answer

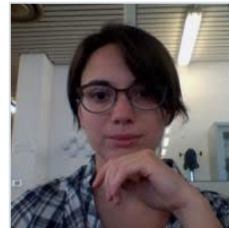
The results are added to the conversation. The model uses them to produce an answer

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BREAK



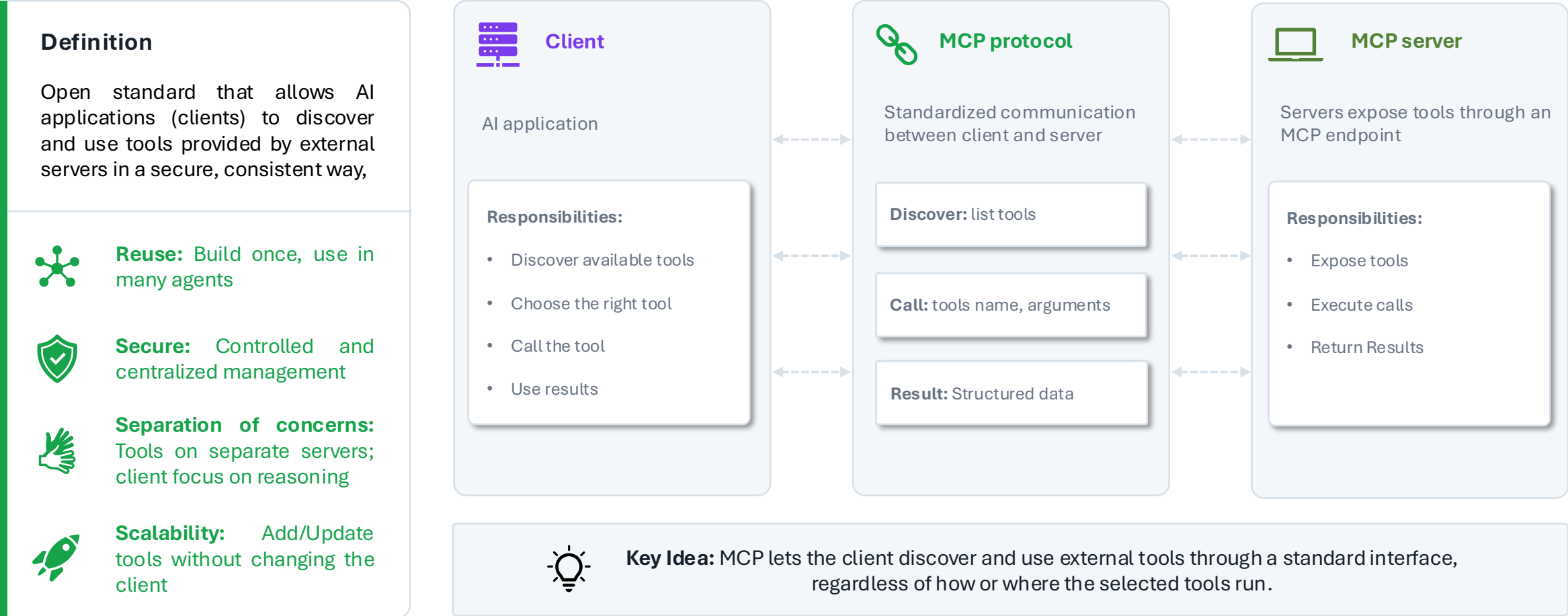
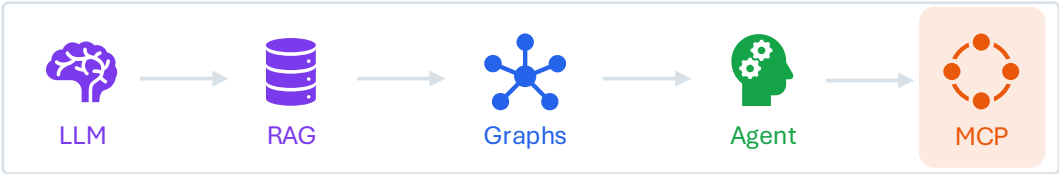
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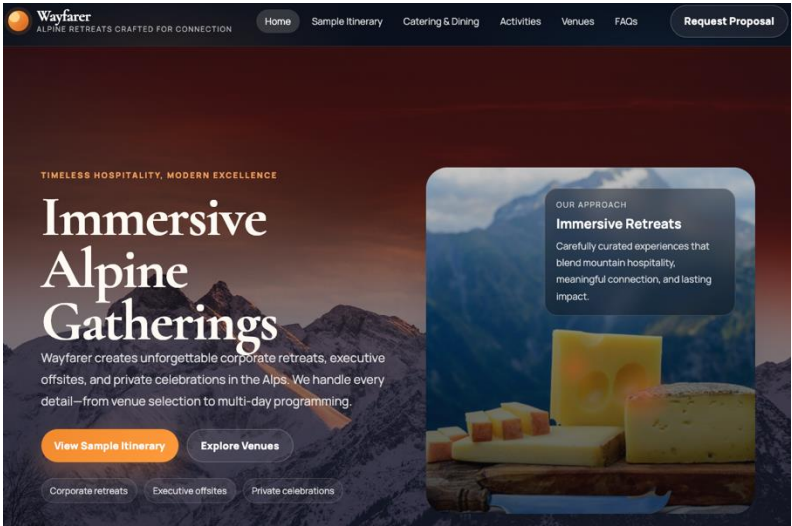
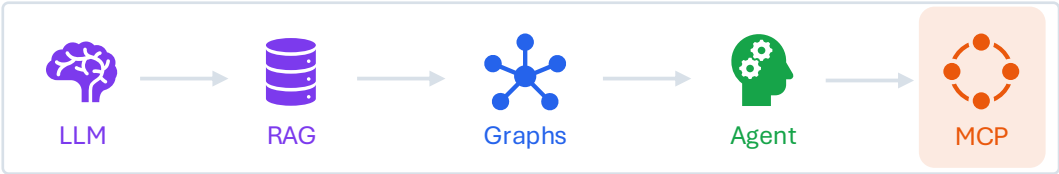
Model Context Protocol (MCP)

Connect Tools Through a Standard Protocol

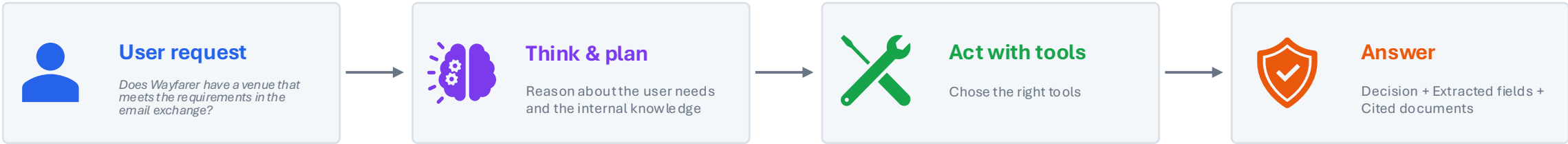


Wayfarer Event Planning Assistant

From business content to evidence-based decisions



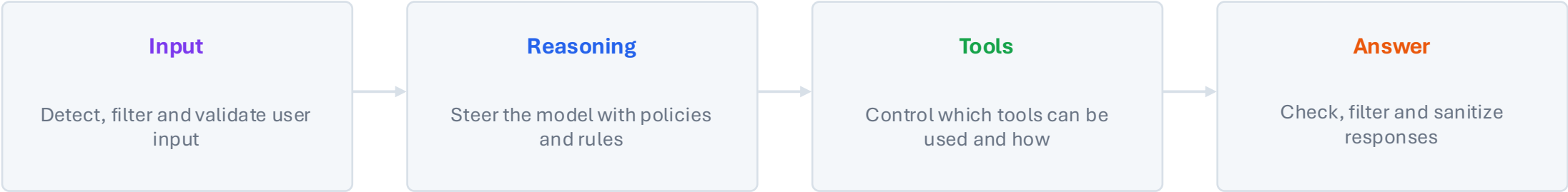
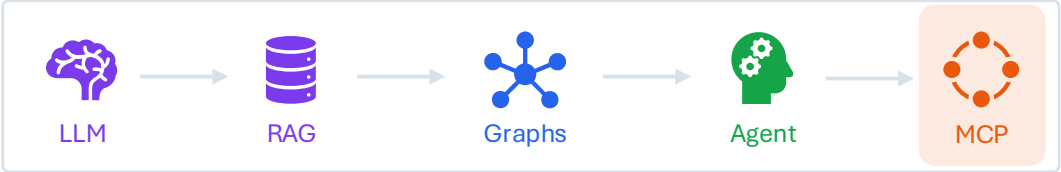
Document pack



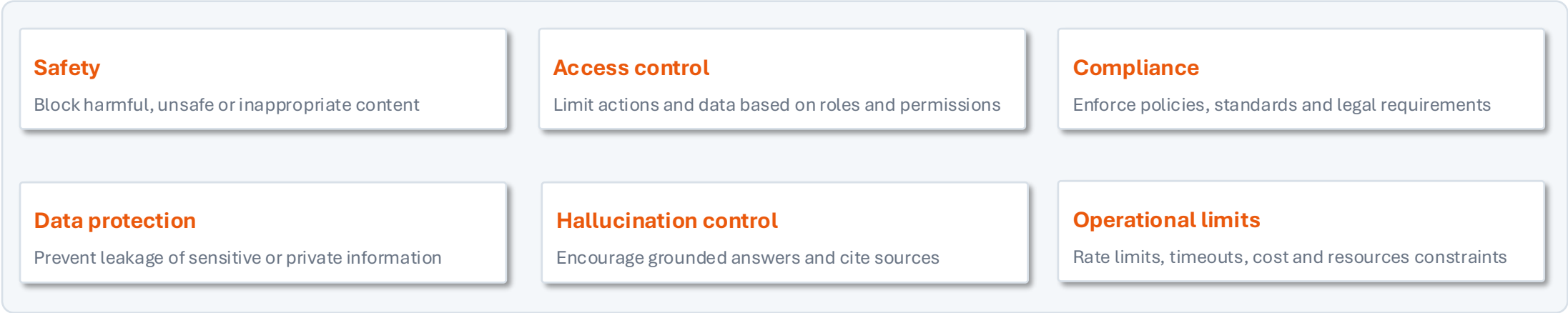
 **Goal:** Combine extraction, retrieval, and useful tools to support reimbursement decisions

Guardrails

Keep the AI assistant safe and reliable



Examples of Guardrails

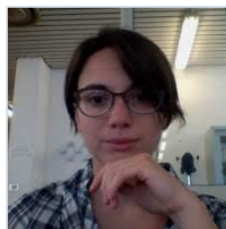


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Thank you !



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